

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

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Problem Statement: Build a multi-agent AI system that helps users manage tasks, schedules, and information by interacting with multiple tools and data sources.

Brief about the idea

NetPulse AI - TelecomOps Multi-Agent Assistant

- A natural-language customer complaint goes in. A structured incident ticket comes out - in under 25 seconds.
- Four ADK sub-agents run in sequence: classify the complaint, investigate BigQuery network events, analyze AlloyDB call records, and write a structured ticket back to AlloyDB. Each agent's output feeds the next via session state.
- NetPulse replaces 10-30 minutes of manual NOC triage across 3 separate systems with one autonomous workflow.
- Built on Google ADK 1.14, MCP Toolbox, BigQuery, AlloyDB AI NL2SQL, and Vertex AI Gemini.

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using ADK, MCP/AlloyDB AI while ensuring Hackathon's core requirements are met?
- What real-world problem does your solution address, and what practical impact does it create for users?
- What is the core approach or workflow behind your solution?

Approach

- Google ADK 1.14 SequentialAgent orchestrates 4 LlmAgent sub-agents. Each owns one tool and feeds the next via output_key session state.
- MCP Toolbox bridges BigQuery and AlloyDB (direct BigQuery MCP fails on Cloud Run).
- Parameterized SQL covers region+window CDR queries in under 2 seconds; AlloyDB AI NL2SQL covers free-form natural-language questions.
- Vertex AI on the global endpoint with 4-attempt model-ladder failover for production reliability.

Real-World Problem

NOC teams query 3 separate systems by hand for every customer complaint. NetPulse compresses 10-30 minutes of triage into 25 seconds by automatically correlating BigQuery network events with AlloyDB CDR evidence.

Core Workflow

Complaint → Classifier → Network Investigator (BigQuery via MCP) → CDR Analyzer (AlloyDB via MCP) → Response Formatter (writes ticket to AlloyDB)

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

What makes NetPulse AI different

- Truly multi-agent. Four specialized sub-agents with structured handoff via session state - not a monolithic agent with many tools.
- Cross-source evidence correlation. Automatically links BigQuery network events with AlloyDB CDRs to surface root cause without human joining.
- Two-tier CDR analysis. Parameterized SQL for region+window CDR queries (sub-2s); AlloyDB AI NL2SQL for free-form natural-language questions. Bounded latency on common patterns, full flexibility on everything else.
- Production-grade failover. 4-attempt Vertex AI model ladder (preview → retry → preview sibling → GA) so the demo never hard-fails under quota pressure.
- Persistent structured output. Every run writes an auditable incident ticket to AlloyDB - queryable, not just chat.

List of features offered by the solution

Core Features

- Classifies complaints into 5 categories: network, billing, hardware, service, general.
- Detects 10 Indonesian regions: Jakarta, Surabaya, Bandung, Medan, Semarang, Yogyakarta, Denpasar, Makassar, Palembang, Balikpapan.
- Investigates network events via 3 BigQuery tools (50,000 events, day-partitioned, region-clustered).
- Analyzes CDRs in two tiers: parameterized SQL for region+window queries (sub-2s) and AlloyDB AI NL2SQL for free-form natural-language questions.
- Writes structured incident tickets to AlloyDB with NOC recommendation.
- Streaming UI with per-agent timeline, model-attempt chips, and customer-impact card.
- Boot-resilient: agent boots even if MCP Toolbox is cold.

Live Deployment

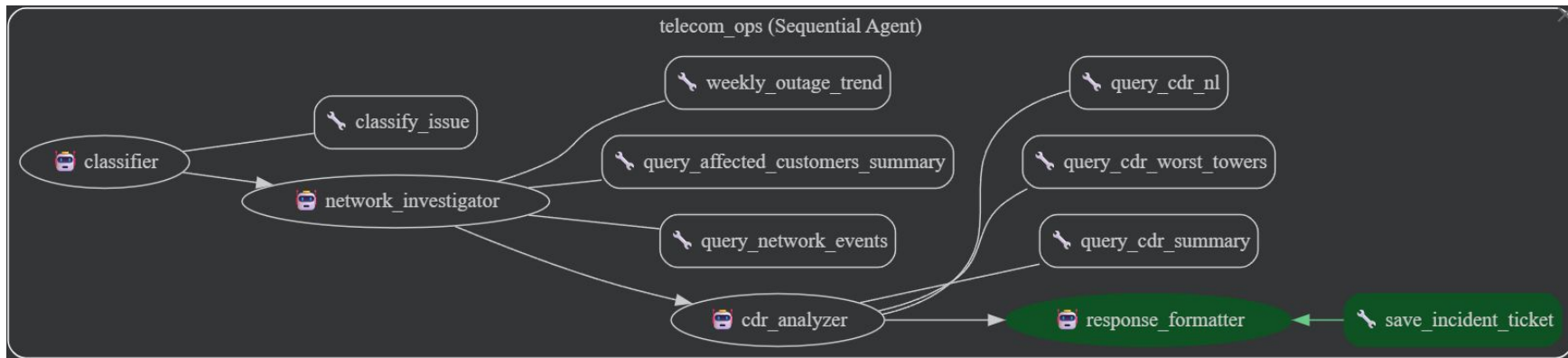
- NetPulse UI: <https://netpulse-ui-486319900424.us-central1.run.app>
- ADK Dev UI: <https://telecom-ops-assistant-486319900424.us-central1.run.app>

Process flow diagram or Use-case diagram

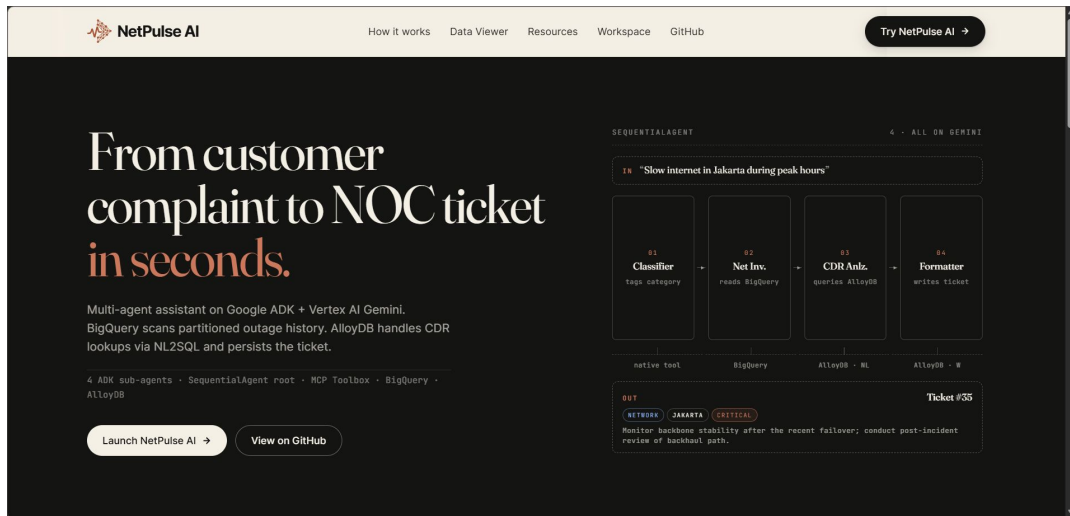
telecom_ops is a Google ADK SequentialAgent orchestrating four LlmAgent sub-agents in fixed order:

1. Classifier - natural-language complaint → structured category and region.
2. Network Investigator - queries BigQuery via MCP Toolbox for events in that region.
3. CDR Analyzer - parameterized SQL for region+window queries, AlloyDB AI NL2SQL for free-form natural-language questions.
4. Response Formatter - writes a structured ticket to AlloyDB and emits the final report.

Each agent's output_key feeds the next via session.state.



Wireframes/Mock diagrams of the proposed solution

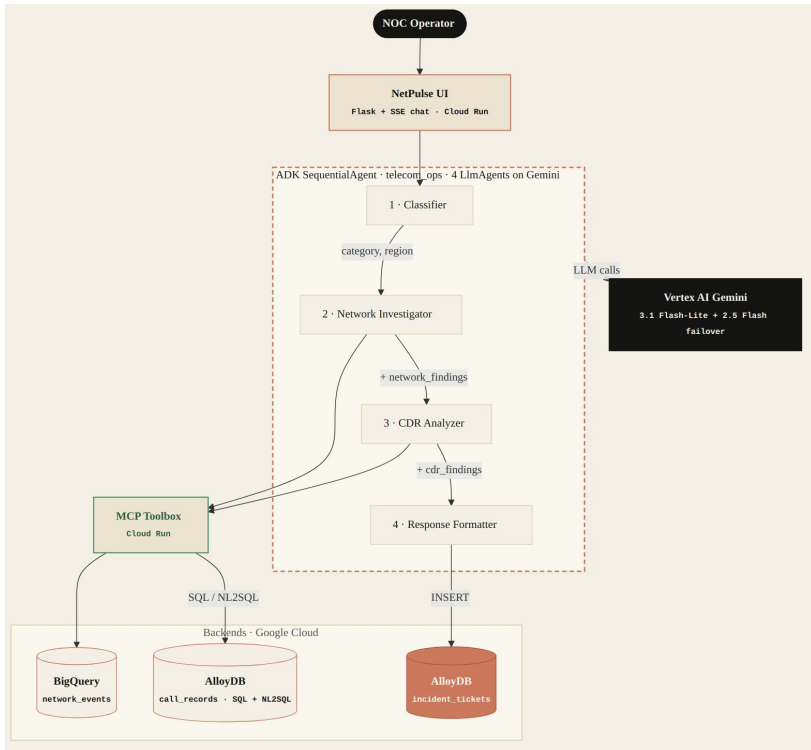


NetPulse AI's UI surface (six pages):

- Landing (/) - hero with "How it works" overview and launch chips that pre-fill the workspace.
- Workspace (/app) - complaint input on top; four streaming agent cards with per-attempt model chips (showing failover hops when they happen); a customer-impact card pulled from BigQuery; a dark Incident Report card at the bottom.
- Network Events (/network-events) - read-only BigQuery viewer with region and severity filters.
- Call Records (/call-records) - read-only AlloyDB viewer over call_records.
- Incident Tickets (/tickets) - read-only AlloyDB viewer over the tickets the response formatter writes.
- Docs (/docs) - in-app technical reference with architecture diagram and stack notes.

Server-Sent Events stream tool calls and text as the SequentialAgent runs, so each card animates as the agent progresses.

Architecture diagram of the proposed solution:



A NOC operator hits one of two Cloud Run frontends - the NetPulse UI (Flask + SSE) or the ADK Dev UI. Both call into the same ADK Runner driving the telecom_ops SequentialAgent.

Each LlmAgent has dedicated tools:

- Classifier → classify_issue (native Python).
- Network Investigator → 3 BigQuery tools via MCP Toolbox.
- CDR Analyzer → 2 parameterized SQL tools (region+window queries) + 1 AlloyDB AI NL2SQL tool (free-form natural-language questions) via MCP Toolbox.
- Response Formatter → save_incident_ticket (writes to AlloyDB).

All four LLM calls go through RegionFailoverGemini: primary gemini-3.1-flash-lite-preview on the global endpoint, with automatic failover to gemini-3-flash-preview and gemini-2.5-flash GA on quota or timeout. Both frontends reach AlloyDB via direct VPC egress.

Technologies/Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Google ADK 1.14

SequentialAgent + LlmAgent orchestration. output_key session-state chain wires the four sub-agents.

Vertex AI Gemini (global endpoint)

Primary gemini-3.1-flash-lite-preview with a 4-attempt model-ladder failover to gemini-3-flash-preview and gemini-2.5-flash GA. Each model has independent quota - sustained 429s on one lane don't imply throttling on the others.

MCP Toolbox for Databases (Cloud Run)

ADK-compatible bridge exposing 6 tools: 3 BigQuery + 3 AlloyDB.

AlloyDB AI NL2SQL (alloydb_ai_nl.execute_nl_query)

Translates free-form natural-language CDR questions into SQL - handles any CDR query the parameterized tools don't express directly (custom joins, weekend-vs-weekday comparisons, duration thresholds). Backed by a structural read-only role (netpulse_nl_reader, SELECT + EXECUTE only) - even if the LLM emits DROP TABLE, the role lacks the privilege.

AlloyDB for PostgreSQL

5,000 CDR rows in call_records; persistent incident_tickets writes. Reached via direct VPC egress.

BigQuery (telecom_network.network_events)

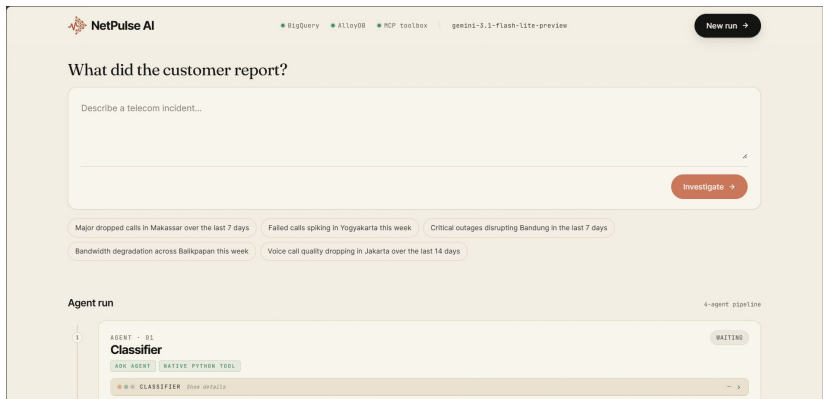
50,000 events across 10 Indonesian cities. Day-partitioned on started_at, clustered by (region, severity).

Snapshots of the prototype

NetPulse AI Custom UI

Top row: Chat workspace (ADK SequentialAgent timeline) and Network Events viewer (BigQuery).

Bottom row: Call Records viewer (AlloyDB read-only) and Incident Tickets viewer (AlloyDB write).



NetPulse AI

BigQuery AlloyDB MCP toolbox genai-3.1-flash-lite-preview New run

What did the customer report?

Describe a telecom incident...

Investigate

Major dropped calls in Makassar over the last 7 days Failed calls spiking in Yogyakarta this week Critical outages disrupting Bandung in the last 7 days

Bandwidth degradation across Balikpapan this week Voice call quality dropping in Jakarta over the last 14 days

Agent run

AGENT - 01

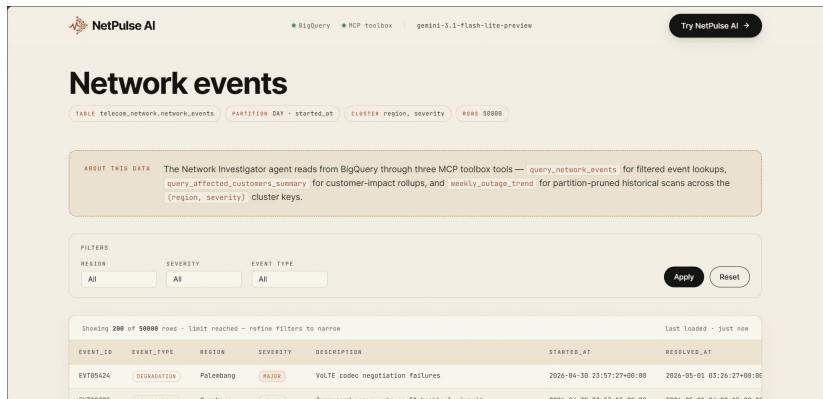
Classifier

ADD AGENT NATIVE PYTHON TOOL

CLASSIFIER Show details

4-agent pipeline

WAITING



NetPulse AI

BigQuery MCP toolbox genai-3.1-flash-lite-preview Try NetPulse AI

Network events

TABLE: telecom_networks.network_events PARTITION: DAY - started_at CLUSTER: region, severity ROWS: 50000

ABOUT THIS DATA The Network Investigator agent reads from BigQuery through three MCP toolbox tools — `query_network_events` for filtered event lookups, `query_affected_customers_summary` for customer-impact rollups, and `weekly_outage_trend` for partition-pruned historical scans across the `(region, severity)` cluster keys.

FILTERS

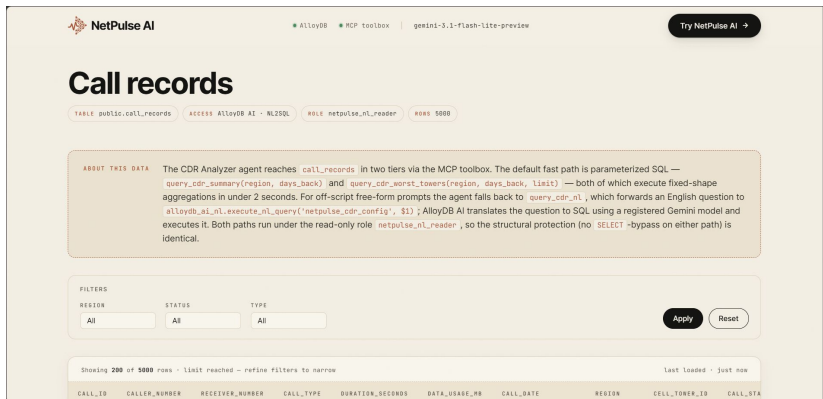
REGION SEVERITY EVENT TYPE

All All All

Apply Reset

Showing 200 of 50000 rows - limit reached - refine filters to narrow Last loaded - just now

EVENT_ID	EVENT_TYPE	REGION	SEVERITY	DESCRIPTION	STARTED_AT	RESOLVES_AT
EVT05424	DEGRADATION	Palembang	MAJOR	VOLTE codec negotiation failures	2024-04-30 22:57:27+00:00	2024-05-02 03:26:27+00:00
EVT00306	DEGRADATION	Surabaya	MAJOR	Increased error rate on P3 backbone circuit	2024-04-30 23:53:15+00:00	2024-05-01 04:08:15+00:00



NetPulse AI

AlloyDB MCP toolbox genai-3.1-flash-lite-preview Try NetPulse AI

Call records

TABLE: public.call_records ACCESS: AlloyDB AI - NL2SQL ROLE: netpulse_nl_reader ROWS: 5000

ABOUT THIS DATA The CDR Analyzer agent reaches `call_records` in two tiers via the MCP toolbox. The default fast path is parameterized SQL — `query_cdr_summary(region, days_back)` and `query_cdr_worst_towers(region, days_back, limit)` — both of which execute fixed-shape aggregations in under 2 seconds. For off-script free-form prompts the agent falls back to `query_cdr_nl`, which forwards an English question to `nl2sql_ai.execute_nl_query('netpulse_cdr_config', $1)`; AlloyDB AI translates the question to SQL using a registered Gemini model and executes it. Both paths run under the read-only role `netpulse_nl_reader`, so the structural protection (no `SELECT` -bypass on either path) is identical.

FILTERS

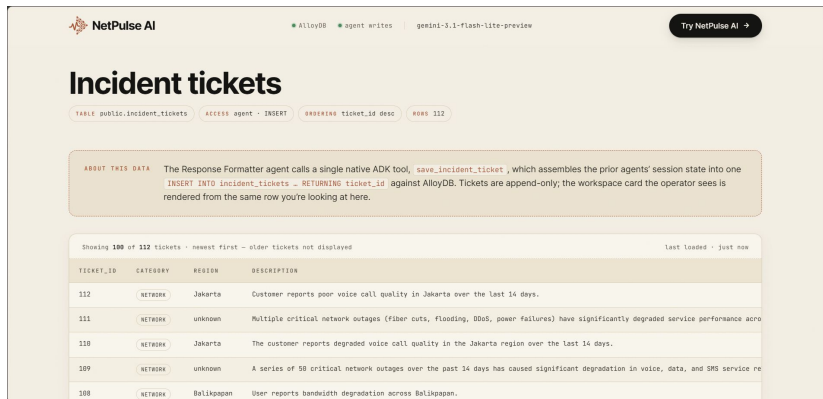
REGION STATUS TYPE

All All All

Apply Reset

Showing 200 of 5000 rows - limit reached - refine filters to narrow Last loaded - just now

CALL_ID	CALLER_NUMBER	RECEIVER_NUMBER	CALL_TYPE	DURATION_SECONDS	DATA_USAGE_MB	CALL_DATE	REGION	CELL_TOWER_ID	CALL_STA
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NetPulse AI

AlloyDB agent writes genai-3.1-flash-lite-preview Try NetPulse AI

Incident tickets

TABLE: public.incident_tickets ACCESS: agent - INSERT ORDERING: ticket_id DESC ROWS: 112

ABOUT THIS DATA The Response Formatter agent calls a single native ADK tool, `save_incident_ticket`, which assembles the prior agents' session state into one `INSERT INTO incident_tickets - RETURNING ticket_id` against AlloyDB. Tickets are append-only; the workspace card the operator sees is rendered from the same row you're looking at here.

Showing 100 of 112 tickets - newest first - older tickets not displayed Last loaded - just now

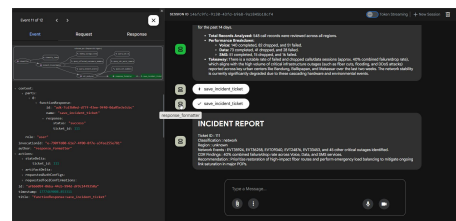
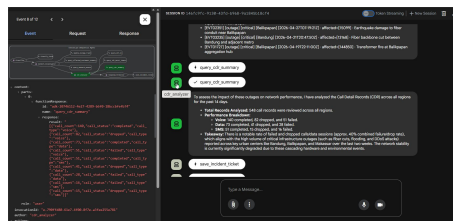
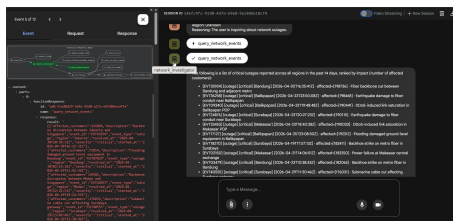
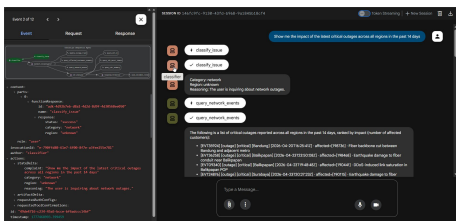
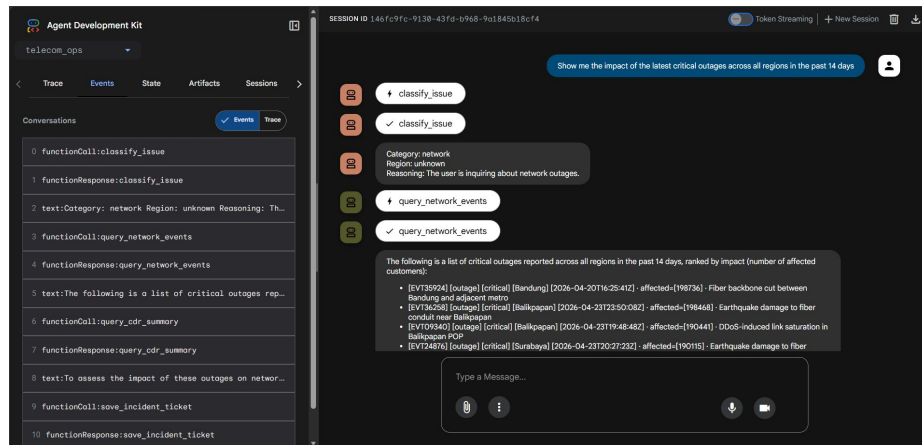
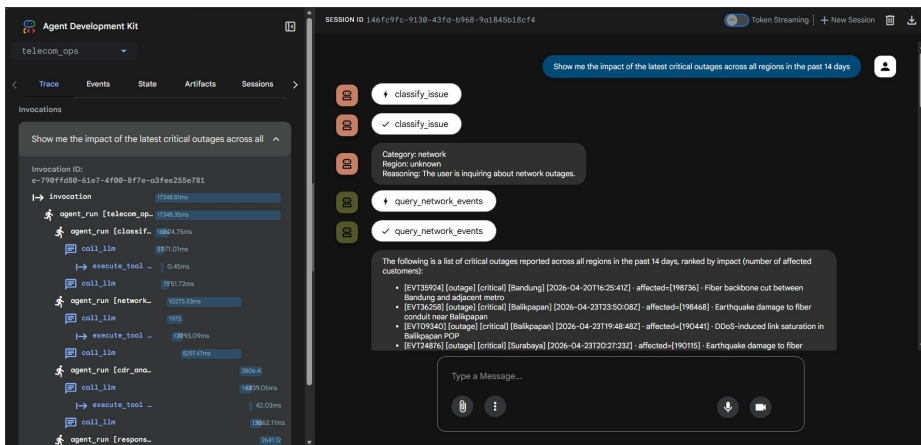
TICKET_ID	CATEGORY	REGION	DESCRIPTION
112	NETWORK	Jakarta	Customer reports poor voice call quality in Jakarta over the last 14 days.
111	NETWORK	unknown	Multiple critical network outages (fiber cuts, flooding, DDoS, power failures) have significantly degraded service performance across
110	NETWORK	Jakarta	The customer reports degraded voice call quality in the Jakarta region over the last 14 days.
109	NETWORK	unknown	A series of 50 critical network outages over the past 14 days has caused significant degradation in voice, data, and SMS service re
108	NETWORK	Balikpapan	User reports bandwidth degradation across Balikpapan.

Snapshots of the prototype

ADK Dev UI

Top: Trace tab (full tool-invocation timeline for one query) and Events tab (streaming sub-agent conversation).

Bottom: the four sub-agents - Classifier, Network Investigator, CDR Analyzer, Response Formatter - each with their tool calls and final output.





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APAC Edition

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Repository: <https://github.com/adityonugroho/hackathon-telecom-ops>

Built with Google ADK 1.14, MCP Toolbox, BigQuery, AlloyDB AI
NL2SQL, and Vertex AI Gemini.

Thank you

